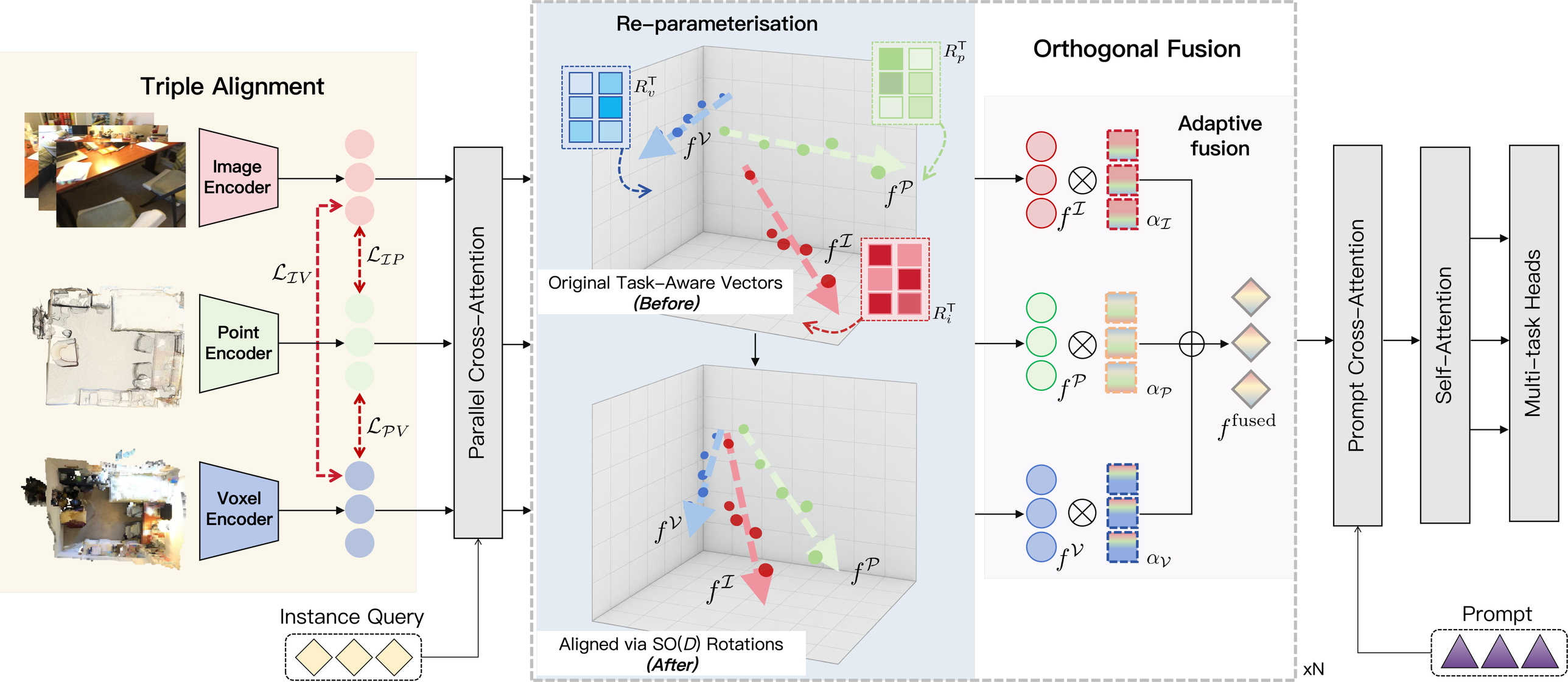

Triple Alignment
Image Encoder
Point Encoder
Voxel Encoder
$\mathcal{L}_{\mathcal{I}\mathcal{P}}$
$\mathcal{L}_{\mathcal{I}\mathcal{V}}$
$\mathcal{L}_{\mathcal{P}\mathcal{V}}$
Parallel Cross-Attention
Instance Query
Re-parameterisation
$R_v^\top$
$R_p^\top$
$R_i^\top$
$f^{\mathcal{V}}$
$f^{\mathcal{P}}$
$f^{\mathcal{I}}$
Original Task-Aware Vectors
(Before)
Aligned via SO(D) Rotations
(After)
Orthogonal Fusion
Adaptive fusion
$\alpha_{\mathcal{I}}$
$\alpha_{\mathcal{P}}$
$\alpha_{\mathcal{V}}$
f^{fused}
Prompt Cross-Attention
Self-Attention
Multi-task Heads
Prompt
xN